LOGISTIC REGRESSION

AIM

To understand the Logistic Regression algorithm by classifying tumours as malignant or benig
for a given data set.

PROGRAM

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# The sigmoid (logistic) function used by Logistic Regression
z = np.linspace(-8, 8, 200)
plt.plot(z, 1 / (1 + np.exp(-z)))
plt.axhline(0.5, color='red', linestyle='--')
plt.xlabel('z')
plt.ylabel('probability')
plt.show()

# Load the data set
data = load_breast_cancer(as_frame=True)
df = data.frame
print("Shape :", df.shape)
print(df.iloc[:, [0, 1, 2, 3, -1]].head())
print(df['target'].value_counts())

features = df.drop('target', axis=1)
label = df['target']
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=23)

# Feature scaling
sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)

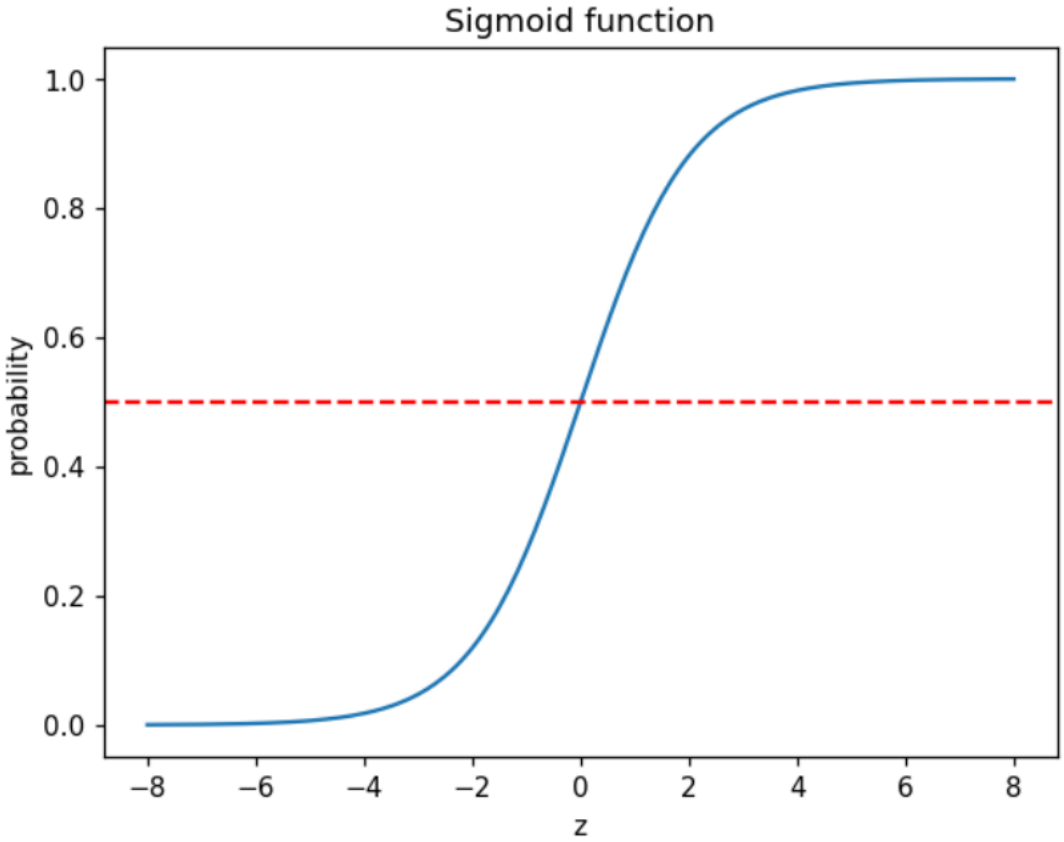
# Training the model
model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)

# Prediction and evaluation
y_pred = model.predict(x_test)
print("Training score :", model.score(x_train, y_train))
print("Testing accuracy:", accuracy_score(y_test, y_pred))
print("Probabilities of first 5 test records [malignant, benign]:")
print(model.predict_proba(x_test[:5]).round(4))
print("Predicted classes :", y_pred[:5])
print("Actual classes  :", y_test.values[:5])

cm = confusion_matrix(y_test, y_pred)
print("\nConfusion matrix:\n", cm)
print("\nClassification report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                    xticklabels=data.target_names, yticklabels=data.target_names)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion matrix')
plt.show()
```

OUTPUT



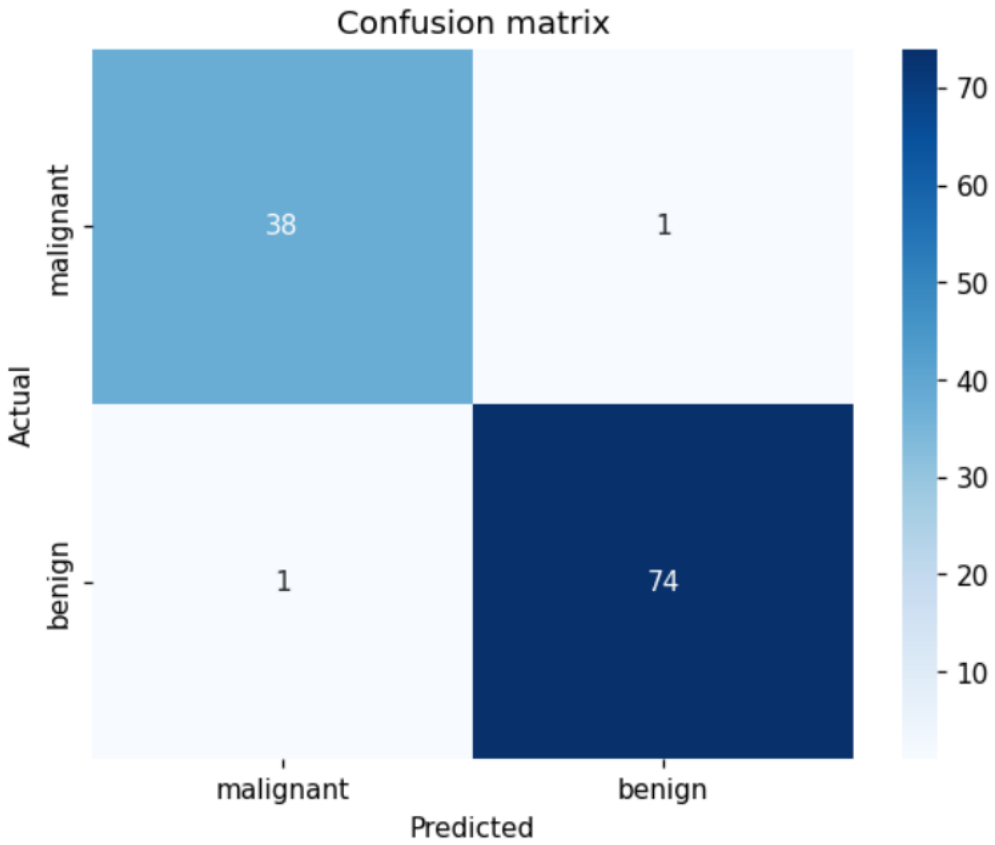

```
Shape : (569, 31)
    mean radius mean texture mean perimeter mean area target
0   17.99     10.38     122.80    1001.0    0
1   20.57     17.77     132.90    1326.0    0
2   19.69     21.25     130.00    1203.0    0
3   11.42     20.38      77.58     386.1    0
4   20.29     14.34     135.10    1297.0    0
Missing values : 0
target
1  357
0  212
Name: count, dtype: int64
Training score : 0.989010989010989
Testing accuracy: 0.9824561403508771
Probabilities of first 5 test records [malignant, benign]:
[[0.0044 0.9956]
 [0.5369 0.4631]
 [1.   0.  ]
 [0.0336 0.9664]
 [0.9983 0.0017]]
Predicted classes : [1 0 0 1 0]
Actual classes  : [1 1 0 1 0]


Confusion matrix:
 [[38 1]
 [ 1 74]]
```

```
Classification report:
                precision  recall f1-score  support

    malignant   0.97   0.97   0.97    39
       benign   0.99   0.99   0.99    75

     accuracy          0.98   114
    macro avg   0.98   0.98   0.98   114
weighted avg   0.98   0.98   0.98   114
```



RESULT

Thus the Logistic Regression model was built and trained on the Breast Cancer data set and it classified the tumours with high accuracy on the test data.